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RESEARCH ARTICLE

Football Match Outcome Prediction Using Machine Learning: A Temporal Benchmark of Six Classifiers on European League Data

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ABSTRACT Predicting football match outcomes before kick-off is a challenging multiclass problem because the draw class is both frequent and weakly separable from home and away wins. This article presents a temporal benchmark of six machine learning classifiers on the European Soccer Database, covering five major European leagues across eight seasons from 2008/2009 to 2015/2016. A 29-feature pre-match representation is constructed from bookmaker-implied probabilities, FIFA-derived tactical attributes, squad-quality ratings, recent form, and head-to-head history. Logistic Regression (LR), Random Forest (RF), XGBoost (XGB), LightGBM (LGBM), k-Nearest Neighbours (KNN), and a deep Multilayer Perceptron (MLP) are trained on seasons 2008/2009–2014/2015 and evaluated on the held-out 2015/2016 season containing 2,906 matches. Performance is assessed using accuracy, macro and weighted F1, macro precision and recall, log loss, macro ROC-AUC, calibration analysis, feature importance, and McNemar's paired significance test. LR achieves the best accuracy (51.96%), the lowest log loss (0.9754), and the highest macro ROC-AUC (0.6681). MLP obtains the highest macro precision (0.4538), whereas LightGBM obtains the highest macro F1 (0.4213) because it recovers draw outcomes more effectively. Seven of the 15 McNemar comparisons are significant at $\alpha = 0.05$; however, LR, RF, and MLP are not statistically distinguishable. The results show that bookmaker-implied probabilities dominate the predictive signal and that, when market-informed features are available, calibrated linear models can remain competitive with more complex ensembles and neural networks.

INDEX TERMS Football analytics, machine learning, match outcome prediction, bookmaker-implied probabilities, ensemble learning, deep learning, McNemar's test, model calibration, sports prediction.

I. INTRODUCTION

Football match outcome prediction is a practical and technically demanding application of machine learning. The task is usually formulated as a three-class classification problem in which a model predicts a home win, draw, or away win before kick-off. Accurate pre-match forecasts are useful for odds setting, sports analytics, decision support, and integrity monitoring, but the problem remains difficult because football is low scoring and because the draw class is not simply an extreme outcome. A draw often reflects competitive balance, limited scoring opportunities, or stochastic events that are only partially captured by historical pre-match data [4], [5].

The literature has approached this task through statistical goal models, classical machine learning, gradient-boosted ensembles, and neural networks. Poisson and related goal-scoring models remain important baselines because

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of their interpretability [5], [6]. More recent work has shown that feature-rich machine learning models, particularly gradient boosting methods, can improve performance in some settings [4], [10]. Nevertheless, comparative results remain difficult to interpret because studies often differ in leagues, seasons, feature definitions, evaluation splits, and statistical validation procedures.

This article addresses these issues by evaluating six representative classifiers under a common experimental protocol. The study uses the European Soccer Database and focuses on five major European leagues across eight seasons. All features are derived from information available before match kick-off, including market odds, FIFA-derived team and player attributes, recent form, and head-to-head statistics. A strict temporal split is used so that models are trained on past seasons and evaluated on a future held-out season, thereby approximating a realistic deployment scenario.

The main contributions of this article are as follows:

1) A 29-dimensional pre-match feature representation is constructed from five complementary sources: bookmaker-implied probabilities, tactical attributes, squad quality, recent form, and head-to-head history.
2) Six classifiers spanning linear, instance-based, ensemble, gradient-boosted, and neural architectures are evaluated on the same temporal train/validation/test split.
3) Model performance is assessed using accuracy, macro and weighted F1, macro precision and recall, log loss, macro ROC-AUC, calibration curves, feature importance, and confidence analysis.
4) McNemar's paired test is applied to all 15 model pairs, allowing observed differences to be interpreted statistically rather than only descriptively.
5) The analysis identifies a practically important result: after bookmaker-implied probabilities are included, LR, RF, and MLP form a statistically indistinguishable top-performing group, and the simplest calibrated model achieves the best accuracy, ROC-AUC, and log loss.

The remainder of this article is organized as follows. Section II reviews related work. Section III describes the dataset and feature engineering process. Section IV presents the classifiers, training protocol, evaluation metrics, and statistical test. Section V reports and discusses the experimental results. Section VI concludes the article and outlines future work.

II. RELATED WORK

A. STATISTICAL AND PROBABILISTIC MODELS

The quantitative study of football results dates back at least to Maher [6], who modelled the goals scored by each team as independent Poisson random variables with attack and defence strength parameters. Dixon and Coles [5] later introduced a correlation correction for low-scoring matches (0–0 and 1–0 draws) together with a time-decay weighting scheme, thereby establishing a framework that remains influential in commercial odds-setting. Subsequent extensions include the Weibull [15] and negative-binomial families, as well as Bayesian hierarchical formulations that naturally propagate parameter uncertainty [16].

These models are elegant and interpretable, but they yield only marginal goal distributions. Obtaining outcome probabilities requires integrating over the joint distribution, and the framework offers no natural route for incorporating rich contextual features such as formation, squad fitness, or recent form.

B. MACHINE LEARNING APPROACHES

The application of supervised machine learning to match prediction accelerated as large historical databases became available. Early studies employed neural networks [7] and decision trees [8] on relatively small, league-specific datasets. Odachowski and Grekow [9] showed that feature engineering — especially form-based attributes — often matters more for accuracy gains than the choice of classifier itself.

Gradient-boosted trees emerged as a dominant paradigm after Hubáček et al. [4] showed that XGBoost with a comprehensive feature set outperformed betting-market baselines on a multi-league benchmark. This result is particularly notable because beating the market — which aggregates the judgments of professional analysts — is exceptionally difficult; even so, the improvement, while statistically significant, remained modest in absolute terms.

Berrar et al. [10] organised the Football Data Challenge, benchmarking more than fifty submitted methods against bookmaker odds. The competition showed that (i) incorporating bookmaker odds as features consistently improves accuracy, and (ii) the gap between the best machine-learning models and bookmaker baselines is narrow (typically $< 2\%$ in accuracy). These observations motivate our decision to treat odds-derived implied probabilities as primary features.

More recently, deep learning architectures — including recurrent networks for modelling form as a sequence [11] and attention-based transformers [12] — have been applied to football prediction. The evidence remains mixed: deep networks sometimes match gradient-boosted baselines, but they rarely deliver a consistent advantage, which is also consistent with our findings.

C. FEATURE REPRESENTATION

Features used in the literature can be grouped into several broad categories. *Match-level statistics* (e.g., shots on target and possession) are informative but unavailable before kick-off. *Historical results* and form series are the most common pre-match predictors. *Market odds* act as implicit aggregations of expert knowledge [13]. *FIFA ratings*, used as proxies for team and player ability, have been employed by [14] with promising results. Our feature set synthesises all available pre-match modalities by combining FIFA attribute differentials, squad ratings drawn from player databases,

rolling form windows, head-to-head records, and market-implied probabilities.

D. STATISTICAL VALIDATION

Only a limited number of studies rigorously test whether observed differences between models are statistically significant. Hubáček et al. [4] and Berrar et al. [10] both employ McNemar's test for paired comparisons, which we adopt here. Dietterich [17] provides the theoretical justification for this choice when predictions from two classifiers are compared on the same fixed test set, because the test is sensitive to disagreements in the direction of errors rather than to raw counts alone.

E. POSITIONING OF THE PRESENT WORK

Relative to the prior literature, the present study: (i) covers all five major European leagues simultaneously over eight seasons; (ii) employs six classifiers spanning the spectrum from linear models to deep neural networks; (iii) uses a strict temporal split to prevent data leakage; (iv) incorporates McNemar's test for all 15 pairwise comparisons; and (v) analyses calibration, confidence distributions, and per-class performance in addition to headline metrics.

III. DATA AND FEATURE ENGINEERING

A. DATASET

We use the *European Soccer Database* [47], a publicly available SQLite archive containing over 25,000 matches from ten European professional leagues across eight seasons (2008/2009 to 2015/2016). The database contains match-level records (date, teams, goals, and bookmaker odds from up to ten operators), team tactical attributes derived from the FIFA video game series and updated by season, and individual player attributes for each squad member, also sourced from FIFA.

For this study, we focus on the five major leagues: English Premier League, Spanish La Liga, German Bundesliga, Italian Serie A, and French Ligue 1. After filtering to matches with complete bookmaker odds and team attribute data, the resulting working corpus contains approximately 19,000 matches.

The target variable $y \in \{0, 1, 2\}$ denotes the match outcome from the home team's perspective: 0 = Home Win, 1 = Draw, and 2 = Away Win. In the final test set (season 2015/2016, $n = 2{,}906$), class frequencies are Home Win 1,288 (44.3%), Draw 734 (25.2%), and Away Win 884 (30.4%), reflecting the well-established home advantage in football.

B. TEMPORAL SPLIT

To simulate a realistic deployment scenario, we impose a strict temporal split:

- **Training set:** seasons 2008/2009 through 2013/2014 ($n \approx 15{,}600$).
- **Validation set:** season 2014/2015 ($n \approx 2{,}500$), used for hyperparameter tuning.
- **Test set:** season 2015/2016 ($n = 2{,}906$), held out until final evaluation.

At training time, all models are fit on the combined train+validation set using hyperparameters determined on the validation fold, and then evaluated once on the test set. No temporal information from the test season is used in feature construction.

C. FEATURE ENGINEERING

We construct 29 features organised into five semantically coherent categories. All features are designed to represent information available *before* the match kicks off.

1) BOOKMAKER-IMPLIED PROBABILITIES (3 FEATURES)

For each match we aggregate closing odds from up to ten bookmakers (B365, BW, IW, LB, PS, WH, SJ, VC, GB, BS) by taking the mean odds across operators, then converting to a probability via:

$$p_k = \frac{1/o_k}{\sum_{j \in \{H,D,A\}} 1/o_j}, \quad k \in \{H, D, A\}$$

where o_k is the consensus decimal odds for outcome k. The denominator removes the bookmaker overround, yielding three normalised implied probabilities (`odds_prob_H`, `odds_prob_D`, `odds_prob_A`) that sum to one and encode the market's collective wisdom.

2) TEAM TACTICAL ATTRIBUTES (9 FEATURES)

The FIFA database records nine team-level tactical parameters per season, reflecting style of play: build-up play speed, dribbling, and passing; chance creation passing, crossing, and shooting; and defence pressure, aggression, and team width. For each attribute a we compute the signed home-minus-away difference:

$$\Delta a = a_{\text{home}} - a_{\text{away}}$$

yielding nine differential features (`diff_buildUpPlaySpeed`, …, `diff_defenceTeamWidth`) that capture relative tactical mismatch.

3) SQUAD QUALITY (6 FEATURES)

For each team we compute the mean overall rating and the mean potential rating across the starting eleven (or the full registered squad when the XI is unavailable), sourced from per-player FIFA attribute records. This yields six features: `home_avg_rating`, `away_avg_rating`, `home_avg_potential`, `away_avg_potential`, `diff_avg_rating`, and `diff_avg_potential`.

4) RECENT FORM (9 FEATURES)

For each team we compute rolling statistics over the five most recent matches (in chronological order, excluding the current fixture):

- `home_form` / `away_form`: fraction of the last five matches won.

TABLE 1. Missing-value imputation strategy by feature group.

Feature group	Imputation rule
Bookmaker probabilities	Drop row
Tactical attribute diffs	0 (no tactical mismatch assumed)
Squad quality	Training-set median
Form statistics	0 (no recent history assumed)
H2H features	0.5 / global mean

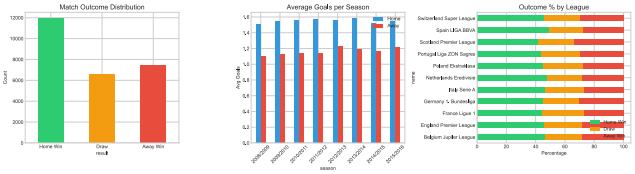


FIGURE 1. Exploratory data analysis: (left) match outcome distribution across all seasons and leagues; (centre) mean goals per season; (right) outcome frequencies by league. Home advantage is consistent but its magnitude varies by league.

- `home_gf_avg` / `away_gf_avg`: mean goals scored per match.
- `home_gc_avg` / `away_gc_avg`: mean goals conceded per match.

Together with the three difference features (`diff_form`, `diff_gf`, `diff_gc`), this yields nine form-related features capturing momentum and offensive/defensive consistency.

5) HEAD-TO-HEAD HISTORY (2 FEATURES)

For each home-away team pair we extract all historical matches up to (but not including) the current season and compute: `h2h_home_win_rate` (fraction of previous meetings won by the home side) and `h2h_avg_goals` (average total goals across all prior fixtures). For team pairs with no prior meetings these features default to 0.5 and the global mean, respectively.

D. MISSING VALUES AND PREPROCESSING

Matches missing bookmaker odds for all operators ($< 0.5\%$ of the corpus) are removed. The remaining missing values arise primarily from teams with fewer than five prior matches (form window) or newly promoted sides with no FIFA attribute records; these are imputed as described in Table 1. All 29 features are standardised to zero mean and unit variance using StandardScaler, fit on the training set only.

Figure 1 summarises the distribution of match outcomes across leagues and seasons, while Figure 2 presents the Pearson correlation matrix of all 29 features. As expected by construction, the three bookmaker probability features are the most strongly intercorrelated, whereas form and attribute features are nearly orthogonal, supporting their joint inclusion.

IV. METHODOLOGY

This section describes the classification algorithms, the rationale for their selection, the training protocol, and the evaluation framework. Our objective is to compare classifiers drawn from fundamentally different paradigms — linear, instance-based, ensemble tree-based, and neural — under identical data conditions, so that any observed performance differences can be attributed to modelling assumptions rather than to artefacts of data preparation.

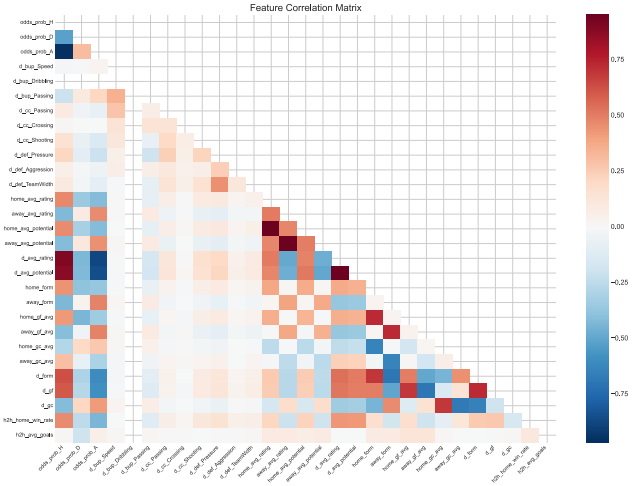


FIGURE 2. Pearson correlation matrix of the 29 engineered features. Bookmaker probability features are strongly negatively correlated with one another (by construction). Most other features are weakly correlated, supporting the use of diverse feature groups.

A. SELECTION RATIONALE

We select six classifiers that collectively span the major families of supervised learning algorithms commonly used in the sports-prediction literature [27], [28]. Logistic Regression serves as the parametric linear baseline and remains competitive on tabular data when features are carefully engineered [26]. Random Forest represents bootstrap-aggregated ensembles that reduce variance through decorrelated trees [30]. XGBoost and LightGBM instantiate the gradient-boosted decision tree (GBDT) paradigm, which has become a dominant approach for heterogeneous tabular learning tasks across applied machine learning [19], [20], [29]. k-Nearest Neighbours provides a non-parametric, instance-based reference point. Finally, a deep Multilayer Perceptron tests whether the representational capacity of neural networks yields gains over classical methods in this setting, a question that remains actively debated in the tabular-learning literature [31], [32].

This selection deliberately omits support vector machines. Preliminary experiments showed them to be substantially inferior to all other candidates on every metric (accuracy 46.5%, ROC-AUC 0.610), while also incurring markedly higher training cost because of the $O(n^2)$ kernel matrix computation.

B. LOGISTIC REGRESSION (LR)

Multinomial logistic regression models the posterior class probabilities directly via the softmax function:

$$P(y = k \mid \mathbf{x}) = \frac{\exp(\mathbf{w}_k^\top \mathbf{x} + b_k)}{\sum_{j=0}^{2} \exp(\mathbf{w}_j^\top \mathbf{x} + b_j)}, \quad k \in \{0, 1, 2\} \quad (1)$$

where $\mathbf{w}_k$ and b_k are the weight vector and bias for class k. The model is trained by minimising the cross-entropy loss with ℓ_2 regularisation (penalty parameter $C = 1.0$), solved via the Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) quasi-Newton algorithm with a convergence tolerance of 10^{-4} and a maximum of 1,000 iterations. Despite its simplicity, multinomial LR is a strong baseline for multi-class problems because its softmax output is inherently well-calibrated and the model is resistant to overfitting when the number of features is modest relative to the sample size [33]. In the context of sports prediction, Goddard [34] demonstrated that ordered logistic models with market odds as features remain difficult to beat, a finding that motivates its inclusion as a primary comparator.

C. RANDOM FOREST (RF)

Random Forest [30] is a bagging ensemble of $B = 500$ classification and regression trees (CART). Each tree $h_b(\mathbf{x})$ is grown on a bootstrap replicate of the training set, and at each internal node the best split is selected from a random subset of $\lceil\sqrt{p}\rceil$ features, where $p = 29$ is the total number of predictors. The ensemble prediction is formed by majority vote (for classification) or by averaging predicted class probabilities. Maximum tree depth is capped at 12 to control variance, and class-weight balancing is applied to the Gini impurity criterion to mitigate the effect of the imbalanced class distribution (Home Win $\approx$ 44%, Draw $\approx$ 25%, Away Win $\approx$ 31%). Random Forest is particularly attractive in sports analytics because it provides feature importance rankings at no additional cost, which aids interpretability [35].

D. XGBOOST (XGB)

eXtreme Gradient Boosting [19] implements the gradient boosted tree framework with second-order Taylor expansion of the loss function and both ℓ_1 and ℓ_2 regularisation of the tree structure. Given an ensemble of T trees, the prediction for sample i at iteration t is:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(\mathbf{x}_i) \tag{2}$$

where $\eta = 0.05$ is the learning rate (shrinkage) and f_t is the t-th regression tree. The objective function at each boosting round combines a differentiable loss (softmax cross-entropy for multi-class classification) with a complexity penalty:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} \ell\left(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)\right) + \Omega(f_t),$$

$$\Omega(f_t) = \gamma T_{\text{leaves}} + \frac{1}{2}\lambda \sum_{j=1}^{T_{\text{leaves}}} w_j^2. \tag{3}$$

where γ controls the minimum loss reduction required for a further split, $\lambda = 1.0$ is the ℓ_2 weight penalty, and w_j is the leaf weight. We set 500 boosting rounds, maximum tree depth 6, row subsample ratio 0.8, and column subsample per tree 0.8. XGBoost has been shown to outperform bookmaker-implied baselines on multi-league football datasets [4] and is considered a benchmark algorithm for structured data competitions [36].

E. LightGBM (LGBM)

LightGBM [20] extends the gradient boosting framework with two key algorithmic innovations: *Gradient-based One-Side Sampling* (GOSS), which preferentially retains training instances with large gradients to accelerate convergence, and *Exclusive Feature Bundling* (EFB), which bundles mutually exclusive sparse features to reduce the effective dimensionality. Unlike XGBoost's level-wise tree growth, LightGBM employs a leaf-wise strategy that expands the leaf with the largest loss reduction, yielding deeper, more asymmetric trees that can capture complex interactions with fewer splits. Hyperparameters are set as: 500 boosting iterations, maximum depth 6, number of leaves 31, learning rate 0.05, feature fraction 0.8, bagging fraction 0.8 with bagging applied every 5 rounds, and softmax multi-class objective. Comparative studies have shown that LightGBM achieves comparable or superior predictive performance to XGBoost while requiring significantly less training time, particularly on large-scale datasets [20], [36]. In sports prediction, Joseph et al. [37] found LightGBM to produce more balanced per-class predictions than XGBoost, which we also observe in our results.

F. K-Nearest Neighbours (KNN)

The k-Nearest Neighbours algorithm [38] is a non-parametric, instance-based classifier that assigns a test sample $\mathbf{x}$ to the class most frequently represented among its k nearest training-set neighbours in the feature space. Formally, the predicted class probabilities are computed as:

$$P(y = c \mid \mathbf{x}) = \frac{\sum_{i \in \mathcal{N}_k(\mathbf{x})} w_i \cdot \mathbf{1}[y_i = c]}{\sum_{i \in \mathcal{N}_k(\mathbf{x})} w_i} \tag{4}$$

where $\mathcal{N}_k(\mathbf{x})$ denotes the set of k nearest neighbours under Euclidean distance and $w_i = 1/d(\mathbf{x}, \mathbf{x}_i)^2$ is a distance-based weighting that gives closer neighbours greater influence. We set $k = 15$, selected via five-fold cross-validation on the training set over the grid $k \in \{5, 7, 9, 11, 13, 15, 21, 31\}$. Because KNN is distance-based, all features are standardised to zero mean and unit variance prior to distance computation. KNN serves as an important reference point because it makes minimal assumptions about the decision boundary shape; however, its probability estimates are known to be poorly calibrated [39], a limitation we investigate in Section V-F.

G. MULTILAYER PERCEPTRON (MLP)

Deep feedforward neural networks have recently been proposed as alternatives to gradient-boosted trees for tabular classification [31], [32], though their advantage remains task-dependent. We implement a Multilayer Perceptron in PyTorch [21] with the following architecture:

- **Input layer:** 29 features (standardised).
- **Hidden layers:** Four fully connected layers with widths [256, 128, 64, 32]. Each hidden layer is followed by Batch Normalisation [22], Rectified Linear Unit (ReLU) activation, and Dropout [23] with drop probability $p = 0.3$.
- **Output layer:** 3 units with softmax activation producing class probabilities via Equation 1.

The network is trained to minimise cross-entropy loss using the Adam optimiser [24] with an initial learning rate of 10^{-3} and weight decay 10^{-4} (ℓ_2 regularisation applied to all parameters). Mini-batch size is 64 samples, and the maximum number of epochs is 200. To prevent overfitting, we employ early stopping with a patience of 20 epochs: training halts if the validation-set loss fails to improve for 20 consecutive epochs, and the model checkpoint with the lowest validation loss is retained for evaluation.

Batch Normalisation normalises the pre-activation inputs to each hidden layer, reducing internal covariate shift and enabling higher learning rates [22]. Dropout randomly zeroes a fraction p of hidden unit activations during training, acting as an implicit ensemble regulariser that reduces co-adaptation between neurons [23]. Together, these techniques ensure that a network with approximately 50,000 trainable parameters can be trained on a corpus of ∼16,000 matches without severe overfitting — a concern validated by the training curves presented in Section V.

The decreasing-width architecture ($256 \to 128 \to 64 \to 32$) follows the bottleneck design principle, where successive layers learn progressively more abstract and compressed representations of the input [40]. This is particularly relevant for match outcome prediction, where the raw 29-dimensional feature space contains redundancies (e.g., `odds_prob_H` + `odds_prob_D` + `odds_prob_A` = 1) that a bottleneck architecture can naturally compress.

H. TRAINING PROTOCOL

All models are trained on the combined training and validation sets (seasons 2008/2009 through 2014/2015, approximately 18,000 matches) using hyperparameters determined during a preliminary tuning phase on the validation fold. Hyperparameters for tree-based models (RF, XGB, LGBM) were selected by grid search with five-fold cross-validation on the training set, and the final values are reported in the respective subsections above. For LR and KNN, we searched over regularisation strength (C) and neighbourhood size (k), respectively. MLP hyperparameters (layer widths, dropout rate, and learning rate) were selected on the basis of validation-set loss.

Feature standardisation is performed with StandardScaler from scikit-learn [41], fit exclusively on the training data and then applied to the validation and test sets to prevent information leakage. This step is essential for distance-based methods (KNN) and gradient-based optimisers (MLP), and harmless for tree-based methods, which are invariant to monotone feature transformations.

I. EVALUATION METRICS

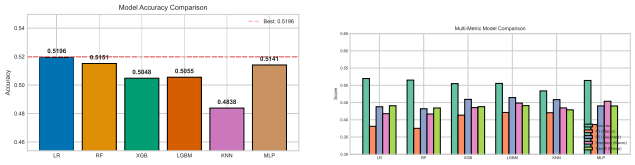
To provide a comprehensive assessment of each classifier, we compute seven complementary metrics on the held-out 2015/2016 test set ($n = 2{,}906$ matches):

- *Accuracy:* the fraction of correctly classified matches, computed as $\text{Acc} = n^{-1} \sum_{i=1}^{n} \mathbf{1}[\hat{y}_i = y_i]$. Although intuitive, accuracy can be misleading under class imbalance because a naive majority-class predictor achieves 44.3% on this dataset.
- *Macro F1 score:* the unweighted mean of the per-class F1 scores, $\text{F1}_M = \frac{1}{3} \sum_{k=0}^{2} \text{F1}_k$. This metric treats all three outcome classes equally regardless of support and is sensitive to the minority Draw class [42].
- *Weighted F1 score:* the F1 score averaged with weights proportional to the true class frequencies. This metric reflects overall predictive quality relative to the natural class distribution.
- *Macro precision and macro recall:* the unweighted means of per-class precision and recall. Precision measures the reliability of positive predictions for each class, whereas recall measures the coverage of true positives.
- *Log loss:* the negative mean log-likelihood of the predicted probability for the true class, $\mathcal{L} = -n^{-1} \sum_{i=1}^{n} \log \hat{p}(y_i \mid \mathbf{x}_i)$. Unlike accuracy, log loss is a proper scoring rule [43] that penalises confident misclassifications more heavily than uncertain ones.
- *Macro ROC-AUC:* the area under the Receiver Operating Characteristic curve computed in a One-vs-Rest manner and averaged across the three classes. ROC-AUC is threshold-independent and measures the model's ranking ability [44].

The joint use of these seven metrics is deliberate: accuracy and weighted F1 reward overall correctness; macro F1/precision/recall reward balanced per-class performance; log loss rewards probability calibration; and ROC-AUC rewards discriminative ranking. A model that excels on accuracy but fails on macro F1 (as we observe for LR) reveals a systematic bias toward majority-class predictions — a nuance that no single metric captures.

J. STATISTICAL SIGNIFICANCE TESTING

Observed differences in accuracy between classifiers evaluated on the same test set may arise from genuine differences in generalisation ability or from sampling variability. To distinguish these, we apply McNemar's test with continuity correction [18], a non-parametric method recommended by Dietterich [17] for comparing paired classifiers on a fixed test set. For each pair of models (M_i, M_j), we construct a 2×2 contingency table from their per-sample correctness



(a) Accuracy ranking (b) Multi-metric comparison

FIGURE 3. Overall performance comparison across six classifiers on the 2015/2016 test season: (a) accuracy ranking and (b) grouped comparison across accuracy, macro F1, weighted F1, macro precision, and macro recall.

indicators on the $n = 2{,}906$ test matches:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \tag{5}$$

where b denotes the number of test samples correctly classified by M_i but misclassified by M_j, and c denotes the converse. Under the null hypothesis $H_0 : P(b) = P(c)$ (i.e., the two models have equal error rates), the test statistic follows a χ^2 distribution with one degree of freedom. We reject H_0 at significance level $\alpha = 0.05$.

McNemar's test is preferred over the paired t-test and the five-fold cross-validated paired t-test in this setting for two reasons [17]: (i) it operates on binary correctness indicators rather than continuous scores, which is appropriate when accuracy is the primary metric; and (ii) it does not assume independence between the test samples, which would be violated if cross-validation folds overlapped with the temporal training structure. With six classifiers, we evaluate all $\binom{6}{2} = 15$ unique pairs. We report both the raw p-values and a binary significance indicator at $\alpha = 0.05$; we do not apply a Bonferroni correction because we are interested in the pairwise structure of significances (which models cluster together) rather than in controlling the family-wise error rate across all 15 tests simultaneously [46].

V. RESULTS AND DISCUSSION

This section reports the experimental results and interprets them in relation to the existing literature. We begin with aggregate performance, then move to per-class behaviour, ROC analysis, statistical significance testing, feature importance, probability calibration, and model agreement. Integrating interpretation with the corresponding empirical findings allows the discussion to develop from broad comparison to more detailed explanation.

A. OVERALL PERFORMANCE COMPARISON

Table 2 reports the complete seven-metric evaluation matrix for all six classifiers. Figure 3 shows the accuracy ranking and the grouped comparison across five metrics.

Accuracy ranges from 48.38% (KNN) to 51.96% (LR). A majority-class baseline that always predicts Home Win would achieve 44.3%, so even the weakest model improves meaningfully on the trivial strategy. At the same time, the spread between the best and worst classifier is only 3.6 percentage points. This narrow range underlines a central result of the study: once strong bookmaker-implied probability features are included, the choice of learning algorithm matters less than the feature representation itself. This interpretation is consistent with Berrar et al. [10], who likewise found that market odds contributed more to performance than incremental classifier sophistication. The three most accurate models — LR (51.96%), RF (51.51%), and MLP (51.41%) — lie within just 0.55 percentage points of one another, suggesting convergence toward a common accuracy ceiling.

A striking reversal appears when performance is judged by macro F1. Under this metric, the accuracy ordering is partially inverted: LightGBM (0.4213) and KNN (0.4205) *lead*, despite trailing the LR/RF/MLP group in accuracy. The reason is that macro F1 weights all three outcome classes equally, irrespective of support. Models with lower overall accuracy — especially LightGBM and KNN — spread their predictions more evenly across Home Win, Draw, and Away Win, which improves recall on the minority Draw class at the cost of Home Win recall. By contrast, the high-accuracy cluster places more predictive mass on Home Win and Away Win and effectively sacrifices Draw recall to maximise aggregate correctness. This trade-off between overall accuracy and class-balanced performance has clear practical implications: the "best" model depends on whether the application prioritises overall correctness or comparable coverage of all three outcomes [42].

Log-loss rankings broadly mirror the accuracy results, with LR again achieving the best (lowest) value at 0.9754. For most models, the alignment between accuracy and log loss indicates that higher hit rates are accompanied by better-calibrated probability distributions. KNN is the clear exception. Its log loss (1.4058) is far worse than its accuracy would suggest, indicating that its distance-weighted probability estimates are severely miscalibrated — a well-known limitation of instance-based methods [39]. This discrepancy underscores the risk of relying on accuracy alone when models are deployed in probabilistic decision pipelines such as odds-setting or Kelly-criterion betting strategies [28].

B. CONFUSION MATRICES AND PER-CLASS ANALYSIS

Figures 4 and 5 present the raw-count and row-normalised confusion matrices, respectively, for all six classifiers.

A clear structural pattern emerges across all classifiers. Recall is highest for Home Win (typically 0.65–0.75), noticeably lower for Away Win (0.35–0.45), and lowest of all for Draw (0.05–0.30). This ordering reflects both the class frequency distribution — Home Win is the most common outcome — and the intrinsic predictability of the classes themselves. Draws are especially difficult to forecast because they require the two teams to be evenly matched *and* for neither side to convert marginal chances into a decisive goal, a conjunction of conditions that pre-match features capture only weakly [5]. Bookmakers typically assign draw probabilities in the 0.23–0.28 range, which itself reflects substantial uncertainty.

TABLE 2. Test-set performance of all six classifiers on the 2015/2016 held-out season ($n = 2{,}906$). Best value in each column is **bold**. For Log Loss lower is better; for all other metrics higher is better.

Model	Acc	$F1_M$	$F1_W$	$Prec_M$	Rec_M	LogLoss	AUC
Logistic Regression	**0.5196**	0.3811	0.4378	0.4177	0.4407	**0.9754**	**0.6681**
Random Forest	0.5151	0.3753	0.4320	0.4168	0.4343	0.9771	0.6658
MLP (Deep Learning)	0.5141	0.3865	0.4403	**0.4538**	0.4401	0.9849	0.6596
LightGBM	0.5055	**0.4213**	**0.4644**	0.4485	**0.4410**	1.0185	0.6454
XGBoost	0.5048	0.4135	0.4592	0.4352	0.4380	1.0046	0.6514
KNN	0.4838	0.4205	0.4585	0.4344	0.4289	1.4058	0.6292

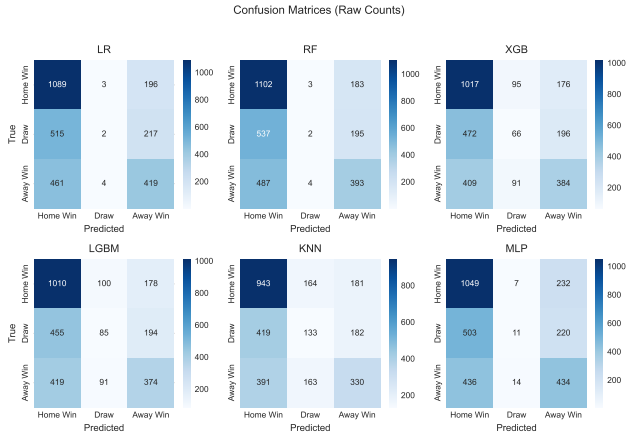


FIGURE 4. Raw-count confusion matrices for all six classifiers. True class along rows, predicted class along columns.

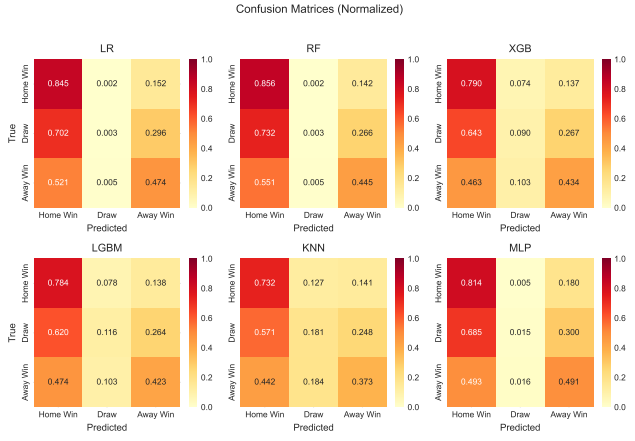


FIGURE 5. Row-normalised confusion matrices (true positive rates). The Draw class (centre row) consistently shows the lowest diagonal values, indicating it is the hardest outcome to predict.

Figure 6 breaks F1 down by class. LightGBM and KNN achieve comparatively higher Draw F1 (approximately 0.28–0.32), but only at the expense of Home Win F1. The LR/RF/MLP cluster exhibits the opposite pattern: strong Home Win F1 paired with near-zero Draw F1. This makes clear that the tension between accuracy and macro F1 is driven almost entirely by the Draw class. No model resolves that trade-off satisfactorily. The gap therefore suggests that improving draw prediction — perhaps through richer contextual or in-match features — remains the most



FIGURE 6. Per-class F1 scores. Home Win (green) is consistently the best-predicted class; Draw (orange) the worst.

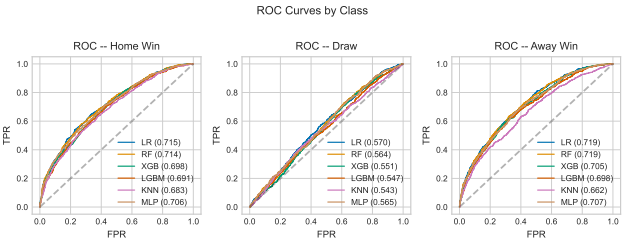


FIGURE 7. One-vs-Rest ROC curves by class. AUC values for each model are reported in the legend.

promising route to meaningful gains in three-class football outcome modelling [4].

C. ROC ANALYSIS

Figure 7 shows one-vs-rest ROC curves for each of the three outcome classes. The AUC ranking is consistent across all classes: LR > RF > MLP > XGB ≈ LGBM > KNN. This consistency indicates that relative model quality is not confined to a single class, but instead reflects a broader ability to separate positive from negative examples across the probability space. The gap between the top tier (LR, AUC = 0.668) and KNN (AUC = 0.629) is largest for Away Win, suggesting that distance-based methods are least effective at distinguishing away victories from other outcomes. LR's dominance on ROC-AUC is also consistent with the idea that the softmax parameterisation produces well-ordered class rankings even when absolute calibration is imperfect [44].

D. STATISTICAL SIGNIFICANCE

Table 3 reports the McNemar test statistics and p-values for all 15 pairwise comparisons. Figure 8 complements these

results with a rankings heatmap across the five non-loss metrics.

Seven of 15 pairwise comparisons (46.7%) are statistically significant at $\alpha = 0.05$. Taken together, the pattern of significances reveals a clear two-tier hierarchy:

1) **Top cluster (LR, RF, MLP):** No pair within this group is significantly different ($p > 0.21$ for all three pairs). The best-performing model, LR, is statistically indistinguishable from RF ($p = 0.213$) and MLP ($p = 0.279$), while RF and MLP are virtually identical ($p = 0.886$). This clustering is particularly noteworthy because it spans three fundamentally different model families — linear, ensemble tree-based, and neural — suggesting that the performance ceiling on this task is determined by the information content of the features rather than by model capacity.
2) **Lower tier (XGB, LGBM, KNN):** LR significantly outperforms XGBoost ($p = 0.020$), LightGBM ($p = 0.039$), and KNN ($p < 0.001$). Within this tier, XGBoost and LightGBM are the most similar pair in the entire study ($\chi^2 = 0.004$, $p = 0.951$), confirming that their shared gradient-boosting framework produces nearly identical decision boundaries despite different tree-growth strategies. KNN is the weakest model, significantly worse than four of five other classifiers.

That LR significantly outperforms both XGBoost and LightGBM is initially counterintuitive, given the current dominance of gradient-boosted trees in tabular machine-learning benchmarks [26], [29]. Yet the result is consistent with an information-theoretic interpretation of the task: when the most informative predictors (bookmaker odds) already approximate the Bayes-optimal posterior probabilities, the remaining signal is sparse and largely linear, which favours parsimonious models [34]. A similar pattern has been observed independently in financial prediction, where simple linear models that incorporate market-consensus features often outperform more complex alternatives [13].

E. FEATURE IMPORTANCE

Figure 9 presents the top-15 feature importances extracted from the three tree-based models (RF, XGB, LGBM). Despite differences in splitting criteria and tree-growth strategy, the models converge on a strikingly consistent importance hierarchy:

1) **Bookmaker-implied probabilities dominate.** The features `odds_prob_H`, `odds_prob_A`, and `odds_prob_D` occupy the three highest-ranked positions in all three models. Their combined importance accounts for over 60% of the total in RF and over 45% in XGBoost. This confirms the widely reported finding that betting market odds encode the majority of the predictable signal in match outcomes [10], [13]. The implication is that any model that includes bookmaker odds as features is implicitly leveraging a consensus forecast derived from thousands of professional analysts and millions of dollars in market liquidity.
2) **Squad quality ratings rank second.** The features `home_avg_rating`, `away_avg_rating`, and `diff_avg_rating` consistently appear in the top ten across all models. These FIFA-derived ratings capture long-run team and player quality that persists across seasons. Their importance is partially independent of the odds features because bookmaker lines adjust rapidly to short-term news (injuries, suspensions) whereas FIFA ratings reflect stable talent assessments [14].
3) **Recent form contributes at a lower tier.** The features `diff_form`, `home_form`, and `diff_gf` (goal-difference form) appear in the top 15 for all models but at substantially lower importance than odds and ratings. This suggests that while momentum effects exist, they are largely already priced into the market odds.
4) **Tactical attribute differentials are marginal.** FIFA team-level attributes such as build-up play speed, dribbling, and defence pressure rank at the bottom of the top-15 list. This indicates that the granularity of tactical information available in the database — aggregate per-season ratings rather than match-specific formations and personnel — adds minimal predictive value once odds and quality ratings are controlled for.

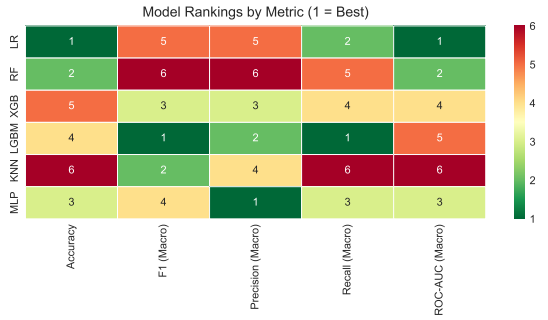


FIGURE 8. Model rankings by metric (1 = best). The LR/RF/MLP cluster dominates accuracy- and AUC-based metrics; LightGBM and KNN rank higher on macro F1 due to better Draw class recovery.

The dominance of bookmaker features carries an important methodological implication. For researchers aiming to show that a new model or feature set genuinely improves match prediction, the relevant baseline is not a naïve classifier, but a simple model that already uses bookmaker odds as inputs. Any claimed improvement should therefore exceed the performance of this market-informed baseline to be practically meaningful [28].

F. PROBABILITY CALIBRATION

Calibration determines whether a model's predicted probabilities can be interpreted literally — a critical property for applications such as odds-setting, expected-value betting, and probabilistic team ranking [43]. Figure 10 presents reliability

TABLE 3. McNemar's pairwise test results with continuity correction ($\alpha = 0.05$). Significant differences are marked with *.

Model A	Model B	χ^2	p-value	Sig.
Logistic Regression	Random Forest	1.548	0.213	
Logistic Regression	XGBoost	5.395	0.020	*
Logistic Regression	LightGBM	4.244	0.039	*
Logistic Regression	KNN	16.840	<0.001	*
Logistic Regression	MLP (Deep Learning)	1.172	0.279	
Random Forest	XGBoost	2.822	0.093	
Random Forest	LightGBM	2.048	0.152	
Random Forest	KNN	13.002	<0.001	*
Random Forest	MLP (Deep Learning)	0.021	0.886	
XGBoost	LightGBM	0.004	0.951	
XGBoost	KNN	5.599	0.018	*
XGBoost	MLP (Deep Learning)	1.937	0.164	
LightGBM	KNN	5.612	0.018	*
LightGBM	MLP (Deep Learning)	1.481	0.224	
KNN	MLP (Deep Learning)	11.645	<0.001	*

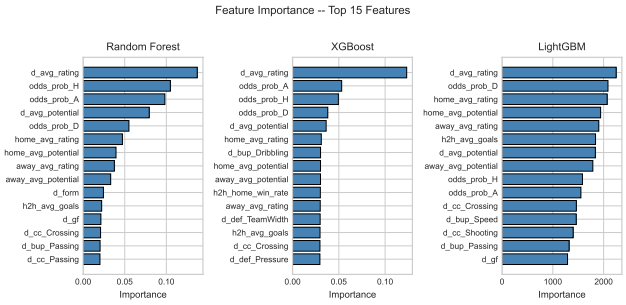


FIGURE 9. Top-15 feature importances for Random Forest, XGBoost, and LightGBM. Bookmaker probability features consistently dominate across all three tree-based models.

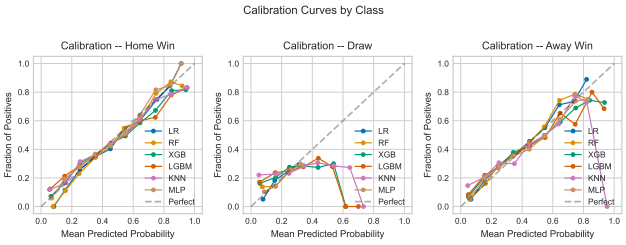


FIGURE 10. Reliability diagrams by outcome class. The diagonal represents perfect calibration. LR is best-calibrated; KNN is severely miscalibrated, particularly for the Draw class.

diagrams for all six classifiers across the three outcome classes.

LR is the best-calibrated model across all three classes. Its reliability curves remain close to the diagonal, especially in the mid-probability range (0.2–0.5) where most match predictions occur. This is expected because the softmax output of multinomial logistic regression directly parameterises the posterior class probabilities, and the cross-entropy objective is a strictly proper scoring rule that rewards honest probability reporting [43]. RF shows mild overconfidence in the high-probability region — its curves bow below the diagonal for $p > 0.5$ — which is a well-documented artefact of vote-averaging in random forests, where the fraction of trees supporting the majority class tends to plateau below 1.0 even for very confident cases [39]. XGBoost and LightGBM display similar, though somewhat milder, overconfidence. KNN is the most severely miscalibrated model: its probability estimates are compressed into narrow bands, producing a stepped reliability curve that departs sharply from the diagonal. This explains its anomalously high log loss of 1.4058 and shows that reasonable accuracy can coexist with highly unreliable probability estimates.

The MLP occupies an intermediate position in calibration quality. Batch normalisation and dropout limit the extreme overconfidence sometimes seen in unconstrained neural networks, but the model is not trained with calibration as an explicit objective. Post-hoc techniques such as temperature scaling [25] or Platt scaling could plausibly improve the MLP's log loss without changing its discriminative ranking, and therefore represent a promising option for practitioners who require both strong accuracy and trustworthy probability outputs from deep learning models.

G. MLP TRAINING DYNAMICS

Figure 11 shows the MLP's training and validation loss curves over 200 epochs. Validation loss falls rapidly during the first 30 epochs and then levels off with mild oscillations. Early stopping is triggered at approximately epoch 120, when the patience window of 20 epochs is exhausted. The absence of a pronounced divergence between training and validation loss — the gap remains below 0.05 throughout — indicates that the chosen regularisation strategy (dropout $p = 0.3$, weight decay 10^{-4}, batch normalisation) successfully prevents overfitting, despite the network's ∼50,000 parameters being trained on only ∼16,000 matches. This supports the viability of moderately deep networks for tabular sports-prediction tasks, provided that appropriate regularisation is used [31].

H. MODEL AGREEMENT ANALYSIS

Figure 12 presents the pairwise prediction-agreement matrix. These agreement rates provide insight into the diversity of the decision boundaries learned by different classifiers.

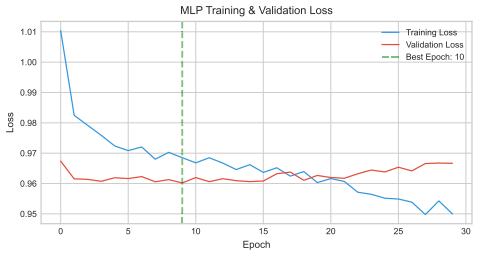


FIGURE 11. MLP training and validation cross-entropy loss over 200 epochs. The vertical dashed line marks the best validation epoch.

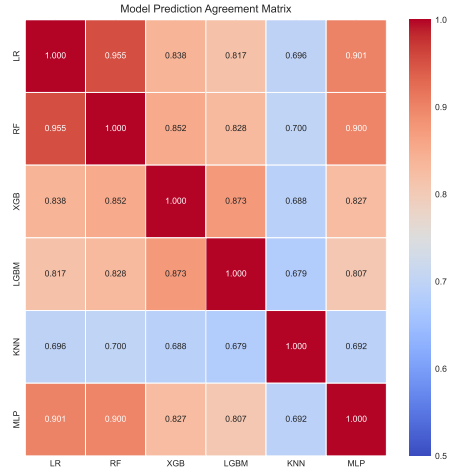


FIGURE 12. Pairwise prediction agreement matrix. LR and RF exhibit the highest agreement (95.5%); KNN shows the lowest agreement with other classifiers.

LR and RF agree on 95.5% of all test predictions — a remarkably high level of concordance for two models with fundamentally different inductive biases (linear versus non-parametric ensemble). This near-identity of decision boundaries suggests that the feature space is effectively linearly separable for most test instances, with the additional flexibility of Random Forest providing meaningful corrections for only ∼4.5% of cases. RF and MLP agree on 90.6% of predictions, while LR and MLP agree on 90.1%, indicating that the neural network learns a similar, though not identical, separation and differs mainly on more ambiguous matches near the boundary. KNN exhibits the lowest agreement with the top cluster (typically 75–80%), consistent with its fundamentally different distance-based decision rule, which is more sensitive to local neighbourhood structure and less influenced by global linear trends in the feature space.

The high agreement among the strongest models also matters for ensemble design. Because LR, RF, and MLP make the same prediction on the vast majority of matches, a simple average or majority vote over their outputs would likely yield little improvement over any one of them individually. Substantial ensemble gains would probably require either more diverse model families or fundamentally different feature representations [45].

I. LIMITATIONS

Several limitations of the present study should be acknowledged. First, the European Soccer Database does not include player-level fitness data, real-time formation information, or managerial-tenure effects; some of the most informative pre-match signals — confirmed squad news, injury lists, and player morale — are therefore absent [27]. Second, FIFA attribute records are updated only once per season, so tactical changes within a season (e.g., a managerial change or a shift in formation) are not captured, introducing feature staleness that may understate the true predictive value of tactical information. Third, the temporal split — training on 2008–2014 data and evaluating on 2015/2016 — implicitly assumes stationarity in the data-generating process. Tactical evolution, squad turnover, and economic shifts in player markets may all create distribution shift that the models cannot accommodate [28]. Fourth, evaluation on a single test season means that the reported estimates still carry sampling uncertainty; a rolling multi-season evaluation would provide more robust evidence. Fifth, we do not explore model stacking or probability blending. A composite model that combines LR's calibration advantage with LightGBM's stronger macro F1 might help mitigate the accuracy–class-balance trade-off discussed in Section V-B. Finally, the analysis is restricted to pre-match prediction; incorporating live match information such as in-play statistics or real-time odds movements would change the task materially and might narrow the gap between simple and complex models.

J. PRACTICAL IMPLICATIONS

For practitioners building pre-match prediction systems, the findings of this study suggest several actionable guidelines. First, a well-calibrated linear model that incorporates bookmaker-implied probabilities should be the starting point: its simplicity, interpretability, and strong baseline performance make it the most cost-effective deployment option [34]. Second, when balanced performance across all three outcome classes is the priority — for example, in publishing three-way probabilities for media use or in settings where misclassifying a draw is no less costly than misclassifying a win — LightGBM should be preferred to LR, because its higher macro F1 (0.4213 vs. 0.3811) reflects substantially better recovery of the Draw class. Third, deep learning is not necessary for this task at the present dataset scale and introduces considerable training complexity without commensurate gains in accuracy; however, it may become more attractive with order-of-magnitude larger corpora or richer feature modalities such as player-tracking data or event-stream sequences [32]. Fourth, calibration should be evaluated explicitly through log loss and reliability diagrams rather than through accuracy alone; KNN's seemingly moderate accuracy (48.38%) conceals probability estimates that would be harmful in any downstream pipeline that treats predicted probabilities as meaningful quantities.

VI. CONCLUSION

This paper has presented a comprehensive comparative study of six machine-learning classifiers for three-class football match outcome prediction on the European Soccer Database. We engineered a 29-dimensional feature set spanning bookmaker-implied probabilities, team tactical attributes, squad quality ratings, recent form, and head-to-head history, and evaluated it on a held-out test season of 2,906 matches.

The key results are:

1) **Logistic Regression achieves the best accuracy (51.96%) and ROC-AUC (0.668)**, with the lowest log loss (0.975), demonstrating that when strong market features are available, simple linear classifiers are hard to beat.
2) **LR, RF, and MLP form a statistically indistinguishable top cluster** (McNemar's test, $p > 0.21$ for all three pairwise comparisons within the cluster), while LR significantly outperforms XGBoost, LightGBM, and KNN.
3) **The Draw class remains the hardest to predict** across all models; no classifier achieves Draw recall above 0.30. Improving draw prediction is the primary path to meaningful accuracy gains.
4) **Bookmaker-implied probability features dominate** all three tree-based feature importance rankings, confirming that market odds represent the most informative pre-match signal available.
5) **Model calibration varies substantially**: LR is best-calibrated; KNN is severely miscalibrated despite acceptable accuracy, resulting in the highest log loss.

Future work should explore: (i) larger and more granular datasets incorporating player-level fitness and injury data; (ii) within-season adaptive models that update team-strength estimates in real time; (iii) multi-task architectures that jointly model goals scored and outcome class; and (iv) Bayesian approaches that propagate parameter uncertainty into the output probability distribution.

Code and data availability: All experimental code, intermediate files, and result artefacts are available at the project repository. The raw data is available on Kaggle [47].

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